

Nicolas DUMINIL

SENIOR SOFTWARE ARCHITECT / DEVELOPER

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Soisy sous Montmorency, France

SKILLS

AI & LLM: Spring AI, LangChain4j, Quarkus LangChain4j, RAG & embeddings, chat memory, tool calling, Model Context Protocol (MCP), structured output, AI agents, Amazon Bedrock, Ollama, Claude Code (skills, agents, plugins, evals).

Languages & Frameworks: Java SE/EE, Spring Boot, Quarkus, Jakarta EE, JPA, Hibernate.

Microservices & SOA: Eclipse MicroProfile, Apache Camel, Apache CXF, REST, SOAP.

DevOps & Cloud: Docker, Kubernetes, Helm, Jenkins, Ansible, AWS (EC2, EKS, Lambda, etc.).

Integration & Messaging: Apache Kafka, ActiveMQ, JMS.

Platforms: JBoss/Wildfly, WebLogic, WebSphere, OpenShift, CloudFoundry, Keycloak.

Databases: Oracle, MySQL, MongoDB.

Tools: Maven, Git, Nexus, Artifactory, Testcontainers, CI/CD pipelines.

AI-enabled Software Architecture & Development (Java SE/EE, Spring Boot, Spring AI, Quarkus, LangChain4j, Kotlin, AWS, OpenShift, Kubernetes, Docker, Jenkins, Helm, GitLab CI/CD, JPA, Hibernate, Apache Camel, CXF, Kafka, JMS)

Communication & Collaboration (Bilingual French/English, fluent in Italian, working knowledge of

PROFILE

- ✓ Multilingual Senior Software Architect with 25+ years of experience in software development and enterprise architecture.
- ✓ Proven expertise in Java SE/EE, SOA, Microservices, and Cloud-native application development.
- ✓ Builds generative AI into Java applications with Spring AI and LangChain4j, on Spring Boot and Quarkus: RAG, tool calling, MCP, structured output and agents. Author of *Java AI Frameworks* (2026).
- ✓ Deep expertise in enterprise systems architecture, integration platforms, and DevOps practices across a variety of sectors including finance, public services, and telecommunications.
- ✓ Proven leadership in designing, building, and modernizing complex systems using technologies like Spring Boot, Quarkus, Kubernetes, and AWS.
- ✓ Author of six published technical books and hundreds of blog articles, with a strong ability to distill complex technical concepts into accessible content for developers, stakeholders, and end users.
- ✓ Proven track record in large-scale projects across Europe in finance, telecom, and public sector.

GENERATIVE AI IN JAVA

2024 - present

- *Java AI Frameworks* (2026): one AI service built three ways (Spring AI on Spring Boot, LangChain4j on Spring Boot, LangChain4j on Quarkus) and graded by one shared conformance test. Covers RAG, chat memory, tools, MCP, vision, audio, structured output, observability, security and agents.
  → leanpub.com/javaaiframeworks · simplex-software.fr/posts-archive/java-ai
- LLM applications with Quarkus and LangChain4j on local models (Ollama), and serverless foundation-model gateways with Amazon Bedrock, Quarkus and AWS Lambda; published as DZone articles.
  → DZone: LLMs with Quarkus & LangChain4j · DZone: Amazon Bedrock with Quarkus
- Claude Code plugin of Spring Boot, Quarkus, Apache Camel and Jakarta EE skills and agents for AI-assisted Java development, its effect measured with Anthropic's evaluation tooling.
  → simplex-software.fr/posts-archive/claude-eval

PROFESSIONAL EXPERIENCES

Founder & Principal Consultant, **Simplex Software** (Paris, since 2004). All engagements below were delivered through Simplex Software.

Senior Java Architect

Thales / Tubize / 2025 - present

- Member of the Thales CDP (Combat Digital Platform) team: software design, development, testing and deployment.
- Stack: Java, Spring Boot, Quarkus, Kubernetes, Keycloak.

Lead Software Architect & Technical Writer

Single Resolution Board (SRB) / Brussels / 2019 - 2024

- Lead Architect & Developer. Designed and developed RDS, a microservice platform for managing European bank data. Stack: Java, Spring Boot, Eclipse MicroProfile, Kafka, Spark, Docker, Kubernetes, AWS.
- Authored detailed system architecture documents, onboarding manuals, and API reference documentation.
- Maintained DevOps documentation for Docker, Kubernetes, and AWS-based infrastructure.

German, Agile, DevOps, cross-functional teams, stakeholder communication)

LINKS

Blog:
https://www.simplex-software.fr

Books:
https://www.simplex-software.fr/books/

Amazon Author Page:
https://shorturl.at/WTBzz

LinkedIn:
https://www.linkedin.com/in/duminilnicolas/

DZone:
https://dzone.com/authors/nicolasduminil

GitHub:
https://github.com/nicolasduminil

LANGUAGES

French

English

Italian

German

Senior Architect & Documentation Lead

Coface, Telefonica, SCOR, Lapeyre, Vlaamse Overheid / 2018 - 2019

- Architected and migrated legacy SOA systems to microservices using Spring Boot, Wildfly Swarm, Apache Camel, Kafka. Cloud deployments on AWS, OpenShift. CI/CD and IaC with Jenkins, Ansible.
- Produced technical specifications, transformation documentation, and knowledge transfer guides for dev teams.

Java EE/DevOps Architect

BNP Paribas Fortis / Brussels / 2015 - 2017

- Designed enterprise-wide digital signature platform (DocuSign). Developed REST/SOAP APIs using Java EE 7. Deployed on WebSphere and Wildfly with containerized DevOps pipeline.
- Contributed to CI/CD and DevOps documentation using Docker, Kubernetes, Jenkins, OpenShift.

Senior Java Architect

La Poste / Paris / 2012 - 2014

- Java EE backend and integration services for large-scale systems. Used Spring Boot, Camel, AWS, Docker, and Kubernetes.

Portal & Enterprise Services Architect

RABO Bank / Utrecht / 2008 - 2011

- Designed and developed Java enterprise applications with IBM WebSphere.
- Authored documentation for internal portal development (JSR-286).
- Developed user and admin guides for deployment and operations.

Earlier experiences highlights

2004–2015: Senior Java Architect roles with ING, ERDF, RABO Bank, Société Générale, Student Loan Company, Deutsche Boerse, EADS, CAP Gemini, and more.
Main activities: SOA/ESB architecture, BPM integration, portal development, system migration, DevOps and CI/CD implementations.

EDUCATION

DEA (Diplôme d'Études Approfondies)
University Pierre et Marie Curie (Paris VI) / Paris

Master's Degree in Electronics & Electrotechnics
ESIEE Engineering School / Noisy-le-Grand

CERTIFICATIONS

- AWS Certified Developer
- Java 11 SE Certified Developer
- Sun Java Certified Programmer
- Microsoft Certified Solution Developer
- Oracle Certified Database Administrator

PUBLISHED BOOKS

From J2EE to AI frameworks, more than twenty years of writing about Java enterprise development and the cloud.



LATEST

Java AI Frameworks

A hands-on comparison of Spring AI and LangChain4j

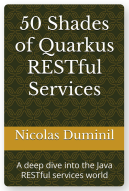
English · Leanpub & Amazon, 2026

Spring AI or LangChain4j? The book answers by building the same application three ways, Spring AI on Spring Boot, LangChain4j on Spring Boot and LangChain4j on Quarkus, with every capability held to one shared conformance test.

What's inside

- Ground answers in your own documents with RAG and tune retrieval
- Add memory and tools, and reach external systems over MCP
- Add vision, audio, structured output, observability and security
- Move from assistants to autonomous agents

Ebook on Leanpub | Buy on Amazon



50 Shades of Quarkus RESTful Services

English · Amazon, 2025

A deep dive into Java RESTful services with Quarkus: MicroProfile, synchronous, asynchronous and reactive services and clients, and their testing.

Buy on Amazon



50 Shades of Java Executors

English · Amazon, 2024

Concurrent programming in Java, with a focus on executors and the latest Java 21 and 23 features.

Paperback | Kindle



AWS Lambda : Développez des micro-services en Java sur la plateforme serverless d'Amazon

French · ENI, 2021

A complete guide to serverless with AWS Lambda: building and deploying enterprise-grade microservices with Quarkus.

Buy on Amazon



AWS : Gérez votre infrastructure sur la plateforme cloud d'Amazon

French · ENI, 2019

A journey through the AWS platform: its configuration, administration and deployment.

Buy on Amazon



J2EE avec JBuilder 7 Enterprise

French · Dunod, 2010

Java Enterprise development, today known as Jakarta EE, formerly Java EE and J2EE.

Buy on Amazon